Supplementary Reproducibility & Evidence Appendix

SAM-assisted auto-labeling and targeted diffusion-augmented learning for warehouse item classification

Evidence snapshot: 12 October 2026 | Source: existing restricted research archive AutoLabelingResearch records and uploaded SAM review archive

A1. Scope and verification boundary

This appendix consolidates saved artifacts without retraining, regenerating images, or rescoring an independent test set. Numerical statements were checked against experiment summaries, comparison CSVs, and a saved human-review archive. Archived source-level records establish internal file-level provenance, not bitwise verification of remote checkpoint binaries or independent certification of test populations.

The 13-class output order is carton_box; crate; plastic_container; pallet; bag_sack; tool; machinery_equipment; electronic_device; vehicle_part; furniture; clothing_textile; raw_material; general_goods. [S01, S07, S12]

A2. Evaluation populations must remain distinct

Population	Recorded size / partition	Permitted interpretation
Original V2 source-group split	3,816 total; 2,685 train / 563 validation / 568 test [S01]	Historical within-project dataset; source-group rule does not override later detected cross-experiment pixel duplicates
Targeted Experiment 9	1,300 total; 1,040 train / 130 validation / 130 test [S07, S08]	Targeted distribution; comparisons valid only on matched images
Hybrid Experiment 10	654 total; 523 train / 65 validation / 66 locked test; only 8 true classes in locked test [S09, S10]	Historical hard-case performance; not a 13-class independent field benchmark
Experiment 11 development	4,109 train and 734 development-validation [S12, S13]	Selection / diagnostic comparisons; not an independently certified external test
SAM quality-control set	39 historical images, 3 per class (uploaded evidence ZIP)	Manual quality-control review only; no independent reference masks

Cross-experiment exposure audit: 5,770 decoded records, 76 exact cross-split pixel-duplicate pairs and 84 near-duplicate candidate pairs awaiting adjudication. There were no missing or unreadable source records in that audit. [S11]

A3. Saved experimental outcomes (do not mix populations)

Model / method	Evaluation	Accuracy	Macro-F1	Interpretive guardrail
Experiment 2 reference	563 V2 validation	62.52%	61.26%	Baseline reference used by Experiment 7 [S05]
Experiment 6 AQCGD	563 V2 validation	63.06%	62.15%	3 added accepted diffusion images; observed, not isolated causal gain [S03-S05]
Experiment 8 frozen router	568 V2 test	56.34%	56.60%	320/568 correct vs Experiment 6 at 319/568; McNemar p=1.0 [S06]
Experiment 9	130 targeted test	80.77%	80.18%	105/130 correct; separate older 568 V2 test = 59.15% accuracy [S08]
Experiment 10	66 eight-class hard-	92.42%	90.83%	61/66; known

Model / method	Evaluation	Accuracy	Macro-F1	Interpretive guardrail
	case test			exposure/coverage limitations [S09-S11]
Experiment 10 on shared dev	734 development-validation	66.21%	65.00%	Compared with Experiment 11 on same 734 [S13]
Experiment 11 Stage 4	734 development-validation	68.66%	68.11%	47 corrected, 29 newly wrong relative to Experiment 10 [S13]
Stage 6F direct resize	734 development-validation	71.25%	70.34%	Padded resize comparison = 69.89% accuracy [S14]
Stage 6G hard negatives	734 development-validation	71.53%	70.75%	Matched control = 71.25% accuracy, 70.45% macro-F1 [S15]
Stage 6H loss modification	734 development-validation	71.53%	70.82%	No accuracy improvement vs matched cross-entropy control [S16]

Stage 5L is separately recorded as an early, two-epoch matched pilot using 11 PROVISIONAL_SYNTHETIC selections. At epoch 2 the real-only arm reached 70.30% accuracy and 69.35% macro-F1, versus 70.16% / 69.33% for diffusion. This does not support a positive diffusion effect. These metrics belong to Stage 5L and must not be combined with Experiment 6 outcomes. [S17, S18]

A4. Diffusion candidate lifecycle and screening

Stage	Saved records	Status / use
Original V2 augmentation	25 accepted crate images; real 2,685 + 25 synthetic = 2,710 training records [S02]	Used in historical model training; distinct from AQCGD
Experiment 6 AQCGD	240 generation attempts; 3 accepted (2 crate, 1 tool); train = 2,713 [S03, S04]	Saved experiment and checkpoint; strong filter rejection
Stage 5F	8 candidates with review_decision=PENDING and training_approved=False [S19]	Not certified as approved synthetic training data
Stage 5K	33 candidates with PENDING_LABEL_QC and training_approved=False [S20]	No confirmed final training approvals in this manifest
Stage 5L	11 PROVISIONAL_SYNTHETIC images [S18]	Used for a limited pilot; not human-approved additions

A5. Checkpoint identity and source-of-hash limitations

Model	SHA-256 stated by archived report	Caution
Experiment 6 (portal audit)	9603b0e6752685d87c8a3148d6368529df75b9893490c0a66c8aa3a03676a03b	Hash from saved local portal reproducibility audit, not recalculated from the remote Drive checkpoint
Experiment 9	397657fbdee35a5309b22a7a6e67168efd9d68855edaf7b4e263c5ae006c47a5	Training summary supplied hash [S07]
Experiment 10	93fb2507db1bd46b3eb7063c0836c7e237c4d6472b7d48ec569a30775a9ceb7d	Saved training metrics and shared-comparison record [S10, S13]
Experiment 11 corrective Stage 4	9ba48d7ae7fe1225fb87169acc5b64b6607f6087f87f81ba3bbae73362cf3777	Saved Stage 4 comparison report [S13]; distinguish from early six-epoch snapshot
Stage 6G active local model	40661fe66915b77fde553dea020d1473d751a42f9a48f490d99aae9520eee30a	SHA from uploaded local active-classifier export; uncalibrated softmax and not independent validation

The first Experiment 11 training_summary.json records six completed epochs and best validation macro-F1 of 66.88%, whereas the later Stage 4 comparison reports a best corrective checkpoint at epoch 22, evaluated on 734

development images (68.66% accuracy / 68.11% macro-F1). These are time-separated research artifacts, not competing final-test measurements. [S12, S13]

A6. SAM quality-control: reviewed decisions vs automatic screen

Saved human-review outcome	Current images
Approved	22
Rejected	1
Correction Required	2
Not Reviewed	14

The uploaded ZIP contains 39 current image records and 26 saved human decision-history entries for 25 distinct reviewed images; two historical approval entries for one image must not be counted as independent specimens. Software fixtures are excluded. Generic legacy reviewer identifiers are not equivalent to authenticated expert adjudication.

Audit caveat: human_review_summary.json reports automatic screening 18 provisional / 18 human-review-required / 3 rejected, whereas the latest run_manifest.json review_status values are 33 ACCEPTED_PROVISIONALLY / 2 REQUIRES_HUMAN_REVIEW / 4 REJECTED_UNUSABLE. These fields reflect different saved status snapshots or semantics and must be reconciled before publishing a single screening breakdown. The explicit human decisions above are the current source for human-review counts.

No independently annotated reference masks or verified human-active annotation-time measurements were established. Approval of a generated mask is not measured IoU, Dice, annotation speedup, or external ground truth. [Uploaded Q1_RESEARCH_EVIDENCE(5).zip]

A7. Reproduction prerequisites and remaining validation

For a reproduction attempt, retain: (i) exact train/validation/test source-image manifests and their image hashes; (ii) 13-class mapping; (iii) preprocessing and architecture configuration; (iv) frozen model checkpoint and its independently recomputed SHA-256; (v) training seed, software versions, scheduler and loss; (vi) split-independent image-level predictions; and (vii) synthetic candidate manifests with review decisions. The archive identifies many of these but this appendix does not claim a complete redistributable code-and-data release.

The original ConvNeXt portal audit records 224×224 direct image resizing, ImageNet RGB normalization means [0.485, 0.456, 0.406] and standard deviations [0.229, 0.224, 0.225], 13 output labels, and a strict checkpoint load test. Do not automatically transfer this audited configuration to every experimental arm without checking arm-specific configs.

Submission-critical unresolved requirements are a genuinely prospective independent 13-class evaluation; expert adjudicated SAM reference masks for overlap metrics; valid human-active timing if efficiency is claimed; and a matched, sufficiently powered diffusion causal ablation if causal benefit is central to the paper. Existing historical tests and small pilots remain valuable descriptive findings.

A8. Permanent evidence index — restricted research archive

Primary workspace: AutoLabelingResearch / experiments. Each [Sxx] identifier below is an exact saved-file location. Links require appropriate restricted research archive permissions; local model paths inside reports are execution history, not public hosting.

S01 V2 original split: 3816 image records, 13-class taxonomy
train_val_test_split_warehouse_item_monitoring_v2_group_safe.json — restricted research archive file

S02 Initial synthetic crate addition and split-safe policy
diffusion_aug_summary_warehouse_item_monitoring_v2_group_safe.json — restricted research archive file

S03 AQCGD training, best epoch and validation metrics
Experiment 6 / training_summary.json — restricted research archive file

S04 AQCGD generator acceptance and rejection accounting
aqcgd_generation_report.json — restricted research archive file

S05 Five-model validation selection comparison
experiment7_validation_model_comparison.csv — restricted research archive file

S06 Frozen-router paired test statistical analysis
step39_final_statistical_analysis.json — restricted research archive file

S07 Experiment 9 training and checkpoint metadata
experiment9_training_summary.json — restricted research archive file

S08 Experiment 9 130-image and 568-image tests
step7_final_test_results.json — restricted research archive file

S09 Experiment 10 split definitions
experiment10_hybrid_split_summary.json — restricted research archive file

S10 Experiment 10 66-image targeted test result
experiment10_final_metrics.json — restricted research archive file

S11 Cross-experiment exact and near-duplicate audit
audit_summary.json — restricted research archive file

S12 Experiment 11 early corrective training summary
experiment11_training_summary.json — restricted research archive file

S13 Experiment 10 vs 11 shared 734-image comparison
stage4_summary.json — restricted research archive file

S14 Stage 6F image-resizing comparison
stage6F_summary.json — restricted research archive file

S15 Stage 6G matched hard-negative comparison
stage6G_summary.json — restricted research archive file

S16 Stage 6H confusion-aware-loss comparison
stage6H_summary.json — restricted research archive file

S17 Stage 5L real-only vs provisional synthetic pilot
stage5L_comparison.csv — restricted research archive file

S18 Stage 5L eleven provisional synthetic selections
pilot_synthetic_selection.csv — restricted research archive file

S19 Stage 5F eight pending synthetic quality reviews
stage5F_verified_approval.csv — restricted research archive file

S20 Stage 5K thirty-three pending synthetic candidates
stage5K_combined_candidates.csv — restricted research archive file

Additional restricted archival evidence includes the historical SAM 39-image review ZIP and the experiment-6 model/preprocessing audit. These sources are not included in this public-deposit candidate, and their original file paths and personal archive locations are kept only in the PRIVATE traceability index. No independent mask ground truth is inferred.

Integrity performed for uploaded SAM ZIP: ZIP CRC scan passed; remote restricted research archive checkpoint bytes were not downloaded and rehashed in this appendix.